

PWM-To-Analogue SIGNAL CONVERTER

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This article presents a simple circuit to convert transistor-transistor logic (TTL) or pulse width modulation (PWM) signal to analogue signal using LM358 operational amplifier. The circuit can be used as an extension shield for Arduino or microcontroller (MCU) or TTL/CMOS signal generator.

PWM-based digital-to-analogue converters (DACs) are often used to produce analogue signals from digital outputs. Usually, solutions for these PWM-based DACs include three main blocks:

1. Digital PWM generator: Mostly, this employs an MCU or Arduino, but can also use a square-wave generator circuit.

2. Analogue integrator: In the simplest case this can be a single RC network. Sometimes, the signal is for driving an electrical motor.

3. Analogue output buffer or amplifier: It provides necessary output voltage, current and power.

Main advantages of PWM DACs include simplicity, low cost, digitally-controlled resolution up to 10 bits (or more), and the possibility to obtain high output current, voltage and power.

Circuit and working

The block diagram of the PWM-to-analogue converter is shown in Fig. 1 and its circuit diagram in Fig. 2. The circuit consists of two parts: input passive RC low-pass filters with three stages and two output stages (OUT1 and OUT2).

Input passive low-pass filters are built around resistors R2, R3 and R4 and capacitors C1, C2 and C3. Cut-off frequencies at -3db for each of the three stages

are calculated according to the formulae given below.

For the first stage,
 $F1 = 1/(6.28 \times R2 \times C1)$

For the second stage,
 $F2 = 1/(6.28 \times R3 \times C2)$

For the third stage,
 $F3 = 1/(6.28 \times R4 \times C3)$

In most cases, $F1 = F2 = F3$ where $R4 = 10 \times R3 = 100 \times R2$ and $R2 \geq 510\text{-ohm}$

Cut-off frequencies at -3db are:
 $F1 = F2 = F3 = 72\text{Hz}$

You may select any appropriate cut-off frequencies for RC filters. Here, we use 5V power supply of the computer or Arduino at CON2 as voltage reference source for the PWM signal.

Output stages are built around two internal operational amplifiers of IC1. First internal operational amplifier (OA1) is formed by pins 2, 3 and 1 of IC1, and the second operational amplifier (OA2) by pins 6, 5 and 7 of IC1.

On closing switch S2, both operational amplifiers work in parallel over the same load at CON4. On opening it, both produce the same voltage over two separate loads. Preferably, the loads should be above 2-kilo-ohm.

If S1 is closed, you can use the power supply of computer V_{cc1} like Arduino at CON2. If S1 is opened, you can use power supply V_{cc2} up to 32V at CON3.

OA1 works as a non-inverting amplifier with the gain ($Av1$) defined by preset VR1 and resistor R7. Minimum gain is +1, and maximum gain is +11. Gain is set according to requirements using VR1, required output voltage and power supply of IC1. It is calculated using the simplified formula given below.

$$Av1 = 1 + VR1/R7$$

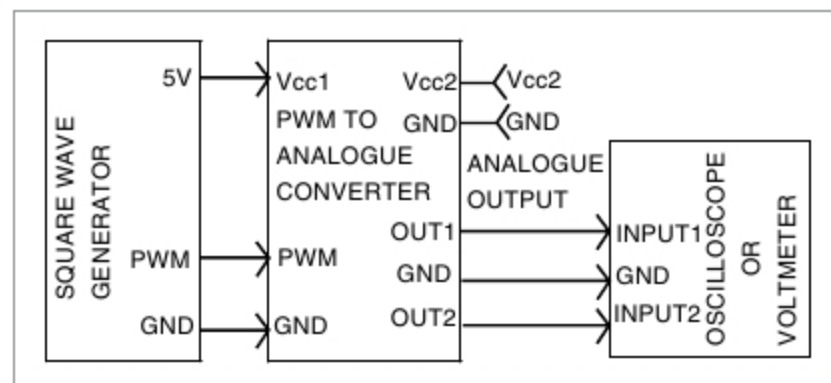


Fig. 1: Block diagram of PWM-to-analogue signal converter

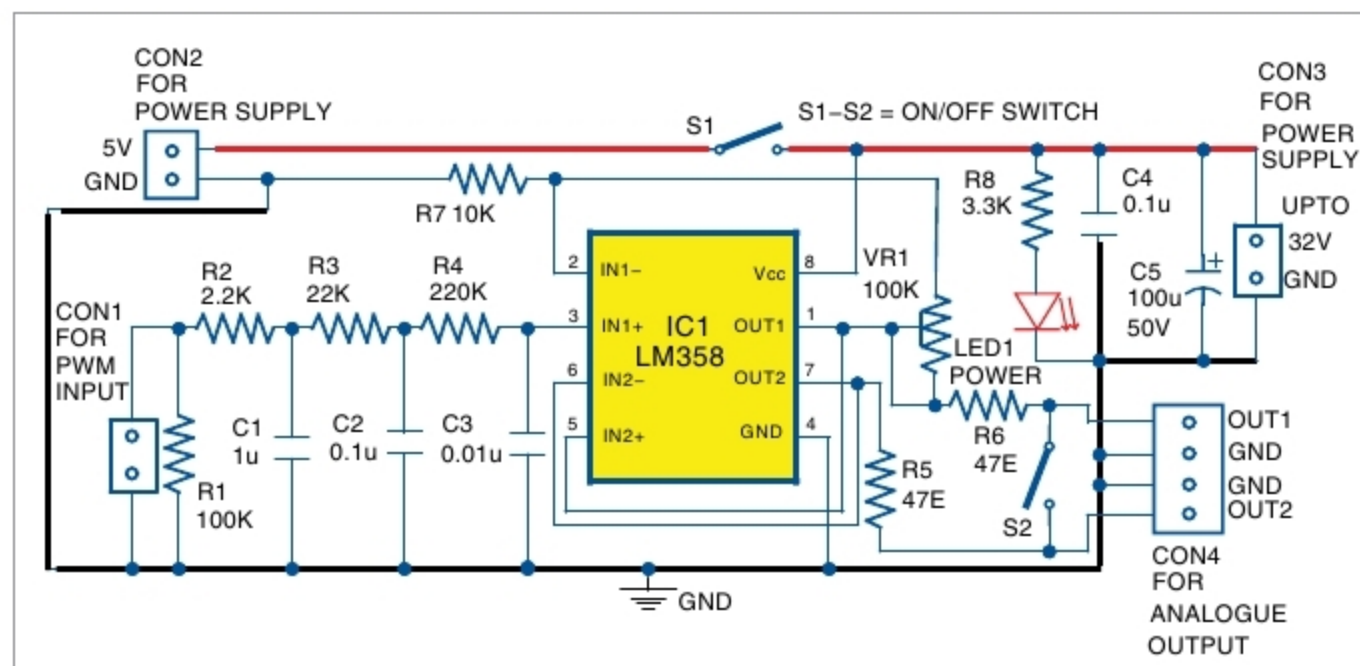


Fig. 2: Circuit diagram of PWM-to-analogue signal converter